Roll No. ....

End-Semester Examination December 2024

Name of the course: B. Tech (ECE) *Semester: VII*

Name of the Paper: CMOS Analog Circuit Design *Paper Code:* **TEC 757**

Time: 3 Hours Maximum Marks: 100

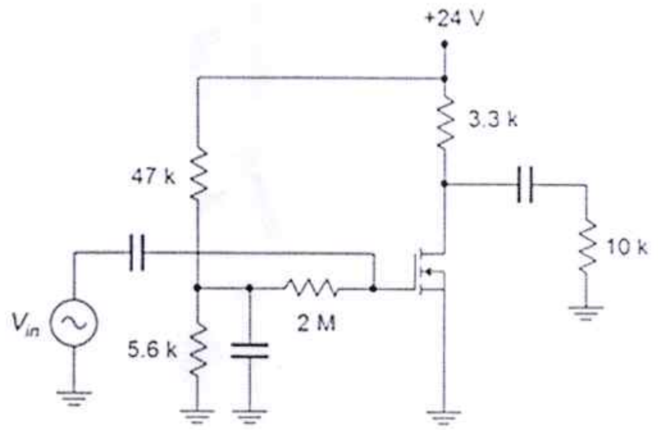
Note:

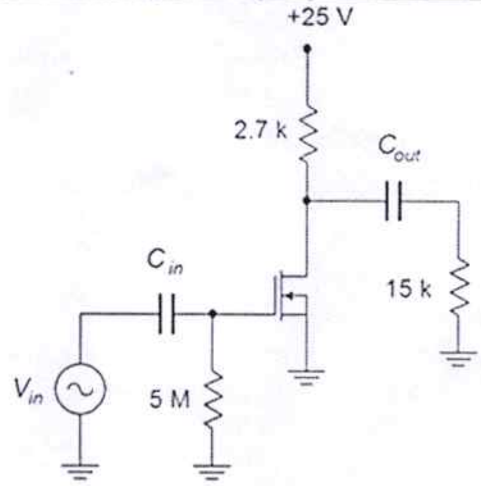
(i) All questions are compulsory.

(ii) Answer any two **questions** among a, b, & c ***in each main question***.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

Q1	**(20 marks)**	
(a)	Explain in brief about frequency response of MOS transistor.	**CO1**
(b)	Explain DC and AC analysis of small signal model MOS transistor.	
(c)	Explain the Basics concepts of MOS transistor.	
Q2	**(20 marks)**	
(a)	Derive expression for gain in diode connected load CS MOSFET amplifier.	**CO2**
(b)	For the circuit of Figure, determine the voltage gain and input impedance. Assume $V_T = 2$ V, $I_{DS} = 50$ mA at $V_{GS} = 5$ V. 	
(c)	For the circuit of Figure, determine the voltage gain and input impedance. Assume $V_{GS} = -0.5$ V and $I_{DSS} = 5$ mA.	

		
Q3	**(20 marks)**	
(a)	Explain the concept of basic differential pair.	CO3
(b)	Explain the concept of Cascode current source.	
(c)	Explain in brief about small and large signal analysis of differential amplifier.	
Q4	**(20 marks)**	
(a)	What do you understand about cascade current mirror?	CO4
(b)	Describe the basic difference between Simple current mirror and Cascode current mirror.	
(c)	Explain in brief about Common source amplifier with complementary load.	
Q5	**(20 marks)**	
(a)	Explain advantages and properties of feedback circuits.	**CO5/6**
(b)	Explain in brief about Simple phase locked loop and its applications.	
(c)	Explain Industrial scope of CMOS analog circuit.	